



Namal University Mianwali

Department of Computer Science

Artificial intelligence

Milestone:02

Submitted by:

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1. Introduction

Measuring biodiversity is crucial to comprehending an environment's ecological health. The purpose of this project is to use computer vision and artificial intelligence to measure biodiversity on a university campus.

The training of deep learning models using plant and animal image datasets is the exclusive focus of this milestone (Milestone 2). In Milestone 3, the campus image test and biodiversity calculation will be carried out.

2. Project Objective

This milestone's goals are:

- To choose appropriate image datasets for animals and plants
- To prepare and preprocess the datasets
- Convolutional Neural Network (CNN) model training
- To assess how well trained models perform

3. Datasets Used

3.1 Plants Dataset – Flowers Recognition Dataset

- **Dataset Name:** Flowers Recognition Dataset

- **Source:** Kaggle
- **Total Images:** ~4,311
- **Number of Classes:** 5

Classes:

- Daisy
- Dandelion
- Rose
- Sunflower
- Tulip

This dataset was used to represent **plant biodiversity**.

3.2 Animals Dataset – Birds-20 Species Image Classification

- **Dataset Name:** Birds-20 Species Image Classification
- **Source:** Kaggle
- **Total Images:** ~3,000+
- **Number of Classes:** 20 bird species

This dataset was chosen due to

- Its manageable size and its clear labeling of bird species.
- Adequate for CNN training with constrained resources
- The biodiversity of animals (birds) is represented by this dataset.

4. Tools & Technologies Used

- Google Colab
- Python
- TensorFlow / Keras
- OpenCV
- Matplotlib
- CNN (Convolutional Neural Network)

5. Data Preprocessing

For **both datasets**, the following preprocessing steps were applied:

- Image resizing to **128 × 128 pixels**
- Normalization of pixel values (0–1)
- Train–Validation split (70% training, 30% validation)
- Data Augmentation:
 - Rotation
 - Zoom
 - Horizontal Flip

Data augmentation was used to improve **model generalization** and reduce overfitting.

6. Model Architecture (Used for Both Datasets)

A **CNN model** was designed using Keras Sequential API.

Model Layers:

- Conv2D (32 filters) + MaxPooling
- Conv2D (64 filters) + MaxPooling
- Conv2D (128 filters) + MaxPooling
- Flatten
- Dense (256 units) + Dropout (0.5)
- Output Dense Layer (Softmax)

Training Parameters:

- **Optimizer:** Adam
- **Loss Function:** Categorical Cross-Entropy
- **Batch Size:** 32
- **Epochs:** 10–15

7. Model Training

7.1 Plant Model Training (Flowers)

- Dataset trained successfully
- Training accuracy increased steadily
- Validation accuracy remained reasonable

Final Results:

- Training Accuracy $\approx 78\%$
- Validation Accuracy $\approx 71\%$

7.2 Animal Model Training (Birds-20)

- Separate CNN model trained on bird images
- Model learned shape, color, and texture features of birds
- Training and validation curves showed stable learning

8. Model Evaluation

For both models, the following evaluation methods were used:

- Accuracy & Loss Curves
- Confusion Matrix
- Classification Report
- Correct vs Incorrect Prediction Visualization

Observations:

- Most predictions were correct
- Some misclassifications occurred between visually similar classes
- No severe overfitting observed

9. Conclusion (Milestone-2)

In this milestone:

- Two datasets (plants & animals) were successfully selected
- CNN models were trained separately for both datasets
- Models achieved reasonable accuracy
- Training phase completed successfully

These trained models will be used in **Milestone-3** to analyze campus images and measure biodiversity.